

## Individual Report

Our project focused on predicting whether a loan application would be approved based on a person's demographics, financial situation, credit history, and loan details. The dataset included features such as age, income, education, work experience, home ownership, credit score, and previous loan defaults, and it had an 80–20 class imbalance. Sam conducted the exploratory data analysis, identifying key patterns, correlations, and imbalance issues. I prepared the dataset by encoding categorical variables, scaling numerical features, balancing the data with SMOTE, and selecting key features using SelectKBest.

Both of us worked together on building and tuning three machine learning models—Logistic Regression, Random Forest, and XGBoost—using pipelines and GridSearchCV. We evaluated the models using accuracy, precision, recall, F1 score, AUC, and confusion matrices. Sam created the conclusion and made final edits to the project, while I designed the presentation for our project. Our best-performing model was XGBoost, which achieved the highest accuracy (0.93), strongest F1 score (0.83), and top ROC-AUC (0.97), making it the most effective and reliable model overall.

### Dataset Reference

Lo, Ta-wei. "Loan Approval Classification Dataset." *Kaggle*, 29 Oct. 2024, [www.kaggle.com/datasets/taweilo/loan-approval-classification-data](https://www.kaggle.com/datasets/taweilo/loan-approval-classification-data).

### 1. Which model did you select, and why?

We selected XGBoost because it achieved the highest test accuracy and ROC-AUC, and it handled nonlinear relationships far better than Logistic Regression while remaining stable across cross-validation folds.

### 2. What features were most important, and do they make sense?

The most influential feature by far was **previous\_loan\_defaults\_on\_file\_Yes**, contributing nearly **80%** of XGBoost's total importance. Other key features—loan-percent-income, credit score, loan amount, and employment experience—also align with real-world loan approval criteria.

### 3. How did model stability or overfitting appear?

XGBoost maintained high and stable test performance (AUC ~0.97) with no major overfitting, despite relying strongly on the default-history feature. Logistic Regression was more stable but less accurate, and cross-validation confirmed XGBoost's consistency.

#### **4. What would you do differently with more time or data?**

We would collect more diverse applicant and credit-behavior data to reduce the model's heavy reliance on default-history and improve fairness. We would also expand the model to support predictions across multiple loan types rather than just a single category.